Chapter 4. Countability and Separation Axioms

First and second countability, regular and normal spaces, Urysohn lemma, Urysohn metrization, Tietze extension

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4.1 The countability axioms

Motivation. Which spaces come from a metric? The answer will be: regular spaces with a countable basis. This chapter builds towards that metrization theorem, starting with the countability axioms, which say that a space can be described by countably many open sets.

X has a **countable basis at** x if some countable collection $\mathcal{B}$ of neighbourhoods of x has every neighbourhood of x containing a member of $\mathcal{B}$; X is **first countable** if this holds at every point. Metric spaces are first countable.

Theorem 4.1 (30.1)

(a) If a sequence of points of A converges to x, then $x \in \overline{A}$; the converse holds if X is first countable.
(b) If $f : X \to Y$ is continuous, it preserves convergent sequences; the converse holds if X is first countable.

Proof. As in Lemma 2.38 and Theorem 2.39: with a countable basis $B_1, B_2, \ldots$ at x, replace it by $B_1 \cap \cdots \cap B_n$ (nested) and pick $x_n \in A \cap B_1 \cap \cdots \cap B_n$. ■

X is **second countable** if its topology has a countable basis. $\mathbb{R}^n$ (rational balls) and $\mathbb{R}^\omega$ (products of rational intervals in finitely many coordinates) are second countable; $\mathbb{R}^\omega$ in the uniform topology is first but not second countable (the sets $\{0,1\}^\omega$ give uncountably many points at mutual distance 1).

Theorem 4.2 (30.2)

Subspaces and countable products of first (resp. second) countable spaces are first (resp. second) countable.

Proof. Intersect a countable basis with the subspace. For $\prod_n X_n$ with countable bases $\mathcal{B}_n$, the products $\prod U_n$ with $U_n \in \mathcal{B}_n$ for finitely many n and $U_n = X_n$ otherwise form a countable basis (a countable union of countable sets, Theorem 1.15); similarly at a point. ■

A is **dense** if $\overline{A} = X$.

Theorem 4.3 (30.3)

If X has a countable basis, then (a) every open covering has a countable subcovering (X is **Lindelöf**), and (b) X has a countable dense subset (X is **separable**).

Proof. (a) For each basis element B_n that lies in some member of the cover, choose such a member A_n; the A_n cover X, since each x lies in some $B_n \subset A$ for A in the cover.

(b) Pick $x_n \in B_n$ (nonempty ones); every nonempty open set contains some B_n, hence some x_n. ■

Example 4.1

$\mathbb{R}_\ell$ is first countable, Lindelöf and separable but not second countable (any basis must contain, for each x, a set B_x with $x = \inf B_x$). $\mathbb{R}_\ell^2$ (the Sorgenfrey plane) is not Lindelöf: the antidiagonal $L = \{x \times (-x)\}$ is closed and each point of L has a neighbourhood $[a,b) \times [-a,d)$ meeting L only there; adding $X - L$ gives a cover with no countable subcover. So products of Lindelöf spaces need not be Lindelöf, and subspaces of Lindelöf spaces need not be Lindelöf. For metrizable spaces, second countable, separable and Lindelöf are equivalent.

4.2 The separation axioms

Motivation. Metric spaces can separate points from closed sets and closed sets from each other by disjoint open sets. The separation axioms (Hausdorff, regular, normal) isolate these properties and measure how far a general space is from being metric.

Assume one-point sets are closed. X is **regular** if for each point x and closed set B not containing x there are disjoint open sets containing x and B; **normal** if each pair of disjoint closed sets has disjoint open neighbourhoods. Normal $\Rightarrow$ regular $\Rightarrow$ Hausdorff.

Lemma 4.4 (31.1)

(a) X is regular iff for each x and neighbourhood U of x there is a neighbourhood V of x with $\overline{V} \subset U$.
(b) X is normal iff for each closed A and open $U \supset A$ there is an open $V \supset A$ with $\overline{V} \subset U$.

Proof. (a) ($\Rightarrow$) Separate x from $B = X - U$ by disjoint $V \ni x$, $W \supset B$; then $\overline{V} \cap B = \emptyset$ since $X - W$ is closed and contains V.

($\Leftarrow$) For $x \notin B$, take $U = X - B$, then V and $X - \overline{V}$ separate.

(b) Same with A for x. ■

Theorem 4.5 (31.2)

(a) Subspaces and products of Hausdorff spaces are Hausdorff.
(b) Subspaces and products of regular spaces are regular.

Proof. (b) Subspace Y: if $x \in Y$ and B closed in Y with $x \notin B$, then $B = \overline{B} \cap Y$ and $x \notin \overline{B}$; separate x from $\overline{B}$ in X and intersect with Y. Product: given $x = (x_\alpha)$ and a basis neighbourhood $\prod U_\alpha$, choose $V_\alpha \ni x_\alpha$ with $\overline{V_\alpha} \subset U_\alpha$ (and $V_\alpha = X_\alpha$ when $U_\alpha = X_\alpha$); then $V = \prod V_\alpha$ has $\overline{V} = \prod \overline{V_\alpha} \subset \prod U_\alpha$ (Theorem 2.30). Apply Lemma 4.4. ■

Example 4.2

(a) $\mathbb{R}_K$ is Hausdorff but not regular: K is closed and $0 \notin K$, but any neighbourhood of 0 has closure meeting every open set containing K.
(b) $\mathbb{R}_\ell$ is normal, but $\mathbb{R}_\ell^2$ is regular (a product of regular spaces) and not normal (Example 4.3). So products of normal spaces need not be normal.

4.3 Normal spaces

Motivation. Normality, separating disjoint closed sets, is the property needed to construct continuous functions, but it is badly behaved under products and subspaces. Here we find the main classes of normal spaces: metric spaces, compact Hausdorff spaces, and regular spaces with a countable basis.

Theorem 4.6 (32.1)

Every regular space with a countable basis is normal.

Proof. Let $\mathcal{B}$ be a countable basis and A, B disjoint closed. Each $a \in A$ has a basis element $U \ni a$ with $\overline{U} \cap B = \emptyset$ (regularity, Lemma 4.4); enumerate those used as $U_1, U_2, \ldots$; they cover A and

their closures miss B. Similarly $V_1, V_2, \dots$ cover B with closures missing A. Put $U_n' = U_n - \bigcup_{i\le n}\overline{V_i}$, $V_n' = V_n - \bigcup_{i\le n}\overline{U_i}$, which are open. $U' = \bigcup U_n' \supset A$ and $V' = \bigcup V_n' \supset B$. If $x \in U_j' \cap V_k'$ with $j \le k$, then $x \in U_j$ but $x \notin \overline{U_j}$ (from V_k'), a contradiction; similarly for $j > k$. ■

Theorem 4.7 (32.2)

Every metrizable space is normal.

Proof. For disjoint closed A, B choose for each $a \in A$ an ϵ_a with $B(a,\epsilon_a)\cap B = \emptyset$, and for each $b \in B$ an ϵ_b with $B(b,\epsilon_b)\cap A = \emptyset$. Let $U = \bigcup_a B(a,\epsilon_a/2)$, $V = \bigcup_b B(b,\epsilon_b/2)$. If $z \in U \cap V$ with $d(a,z) < \epsilon_a/2$, $d(b,z) < \epsilon_b/2$, then $d(a,b) < (\epsilon_a+\epsilon_b)/2 \le \max(\epsilon_a,\epsilon_b)$, contradicting the choice of radii. ■

Theorem 4.8 (32.3)

Every compact Hausdorff space is normal.

Proof. Regularity: Lemma 3.18 with the compact closed set B. Normality: for disjoint closed A, B, choose for each $a \in A$ disjoint $U_a \ni a$, $V_a \supset B$; finitely many U_{a_i} cover A; take $U = \bigcup U_{a_i}$, $V = \bigcap V_{a_i}$. ■

Theorem 4.9 (32.4)

Every well-ordered set is normal in the order topology.

Proof. Every interval $(x, y]$ is open (it equals (x, y^+) with y^+ the successor of y, or a ray if y is largest). Let A, B be disjoint closed, first not containing the smallest element a_0. For $a \in A$ choose $(x_a, a]$ missing B, and for $b \in B$ choose $(y_b, b]$ missing A. $U = \bigcup(x_a, a]$, $V = \bigcup(y_b, b]$ are disjoint: if $z \in (x_a, a]\cap(y_b, b]$ with $a < b$, then $y_b < z \le a < b$, so $a \in (y_b, b]$, which misses A. If $a_0 \in A$, note $\{a_0\}$ is open and closed; apply the argument to $A - \{a_0\}$ and add $\{a_0\}$ to U. ■

Example 4.3

(a) The Sorgenfrey plane $\mathbb{R}_\ell^2$ is not normal. With L the antidiagonal, every subset $A \subset L$ is closed, as is $L - A$. If each pair A, $L - A$ had disjoint open neighbourhoods $U_A \supset A$, $V_A \supset L - A$, then $A \mapsto U_A \cap D$ (D the rational points) would be injective from $\mathcal{P}(L)$ into $\mathcal{P}(D)$, which has the cardinality of $\mathcal{P}(\mathbb{Z}_+)$; but L has the cardinality of $\mathbb{R}$, and $\mathcal{P}(\mathbb{R})$ is strictly larger (Theorem 1.18).
(b) $S_\Omega \times \overline{S}_\Omega$ is not normal although both factors are (Theorem 4.9): the diagonal $A = \{x \times x\}$ and $B = S_\Omega \times \{\Omega\}$ are disjoint closed sets that cannot be separated. Since $\overline{S}_\Omega \times \overline{S}_\Omega$ is compact Hausdorff, hence normal, normality is not hereditary.
(c) $\mathbb{R}^J$ for uncountable J is not normal.

4.4 The Urysohn lemma

Motivation. In a metric space, $d(x,A)/(d(x,A)+d(x,B))$ is a continuous function that is 0 on A and 1 on B. Urysohn's lemma produces such a function in any normal space, without a metric. It is the first deep theorem of the course and the engine of both the metrization and the extension theorems.

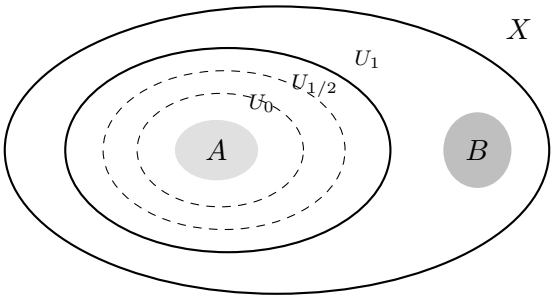


Figure. *Urysohn's construction: open sets U_r, one for each rational $r \in [0,1]$, with $A \subset U_0$, $\overline{U_r} \subset U_s$ for $r < s$, and $U_1 = X - B$. Then $f(x) = \inf\{r : x \in U_r\}$ is continuous, 0 on A and 1 on B.*

Theorem 4.10 (33.1, Urysohn lemma)

Let X be normal and A, B disjoint closed subsets. For any $[a,b] \subset \mathbb{R}$ there is a continuous $f : X \to [a,b]$ with $f = a$ on A and $f = b$ on B.

Proof. It suffices to treat $[0,1]$.

Step 1. let P be the rationals in $[0,1]$, enumerated with $1, 0$ first. Define open U_p $(p \in P)$ such that $p < q \Rightarrow \overline{U_p} \subset U_q$: put $U_1 = X - B$; by Lemma 4.4(b) choose $U_0 \supset A$ with $\overline{U_0} \subset U_1$. If U_p is defined for the first n rationals P_n with the property, let r be the next; $P_n \cup \{r\}$ is finite and ordered, r has an immediate predecessor p and successor q in it, and normality gives U_r with $\overline{U_p} \subset U_r \subset \overline{U_r} \subset U_q$. By induction (recursive definition) all U_p are defined.

Step 2. extend to all rationals p by $U_p = \emptyset$ for $p < 0$ and $U_p = X$ for $p > 1$; the property persists.

Step 3. let $\mathbb{Q}(x) = \{p : x \in U_p\}$, which contains no $p < 0$ and every $p > 1$, and $f(x) = \inf \mathbb{Q}(x) \in [0,1]$. On A, $f = 0$; on B, $x \notin U_p$ for $p \le 1$, so $f = 1$.

Step 4. (i) $x \in \overline{U_r} \Rightarrow f(x) \le r$ (then $x \in U_s$ for all rational $s > r$);

(ii) $x \notin U_r \Rightarrow f(x) \ge r$ (then $x \notin U_s$ for $s < r$).

Step 5. continuity: given x_0 and $(c,d) \ni f(x_0)$, choose rationals $c < p < f(x_0) < q < d$. Then $U = U_q - \overline{U_p}$ is an open neighbourhood of x_0 (by (i), (ii) contrapositives), and for $x \in U$: $x \in U_q \subset \overline{U_q}$ gives $f(x) \le q$; $x \notin \overline{U_p} \supset U_p$ gives $f(x) \ge p$. So $f(U) \subset [p,q] \subset (c,d)$. ∎

A, B are **separated by a continuous function** if there is $f : X \to [0,1]$ with $f(A) = \{0\}$, $f(B) = \{1\}$. A space in which one-point sets are closed and each point x_0 and closed set $A \not\ni x_0$ are so separated is **completely regular**. Normal ⇒ completely regular ⇒ regular (if $f(x_0) = 0$, $f(A) = 1$, the sets $f^{-1}([0, \frac{1}{2}))$, $f^{-1}((\frac{1}{2}, 1])$ separate).

Theorem 4.11 (33.2)

Subspaces and products of completely regular spaces are completely regular.

Proof. Subspace: restrict the function separating x_0 from $\overline{A}$. Product: given $x = (x_\alpha)$ and a basis neighbourhood $\prod U_\alpha$ with $U_\alpha \ne X_\alpha$ only for $\alpha_1, \dots, \alpha_n$, take $f_i : X_{\alpha_i} \to [0,1]$ with $f_i(x_{\alpha_i}) = 1$ and $f_i = 0$ off U_{α_i}; then $f = \prod_i f_i \circ \pi_{\alpha_i}$ is 1 at x and 0 off $\prod U_\alpha$. ∎

So $\mathbb{R}^2_\ell$ and $S_\Omega \times \overline{S}_\Omega$ are completely regular but not normal.

4.5 The Urysohn metrization theorem

Motivation. Combining Urysohn functions with the countable basis, we embed the space in $\mathbb{R}^\omega$, which is metrizable. This answers the metrization question for second countable spaces: they are

metrizable exactly when they are regular.

Theorem 4.12 (34.1, Urysohn metrization theorem)

Every regular space with a countable basis is metrizable.

Proof.

Step 1. there is a countable family $f_n : X \to [0,1]$ such that for each x_0 and neighbourhood U of x_0 some f_n is positive at x_0 and vanishes off U. Indeed X is normal (Theorem 4.6); for each pair (n,m) of indices with $\overline{B_n} \subset B_m$ (B_i the countable basis), Urysohn gives $g_{n,m} = 1$ on $\overline{B_n}$, 0 off B_m. Given $x_0 \in U$, choose B_m with $x_0 \in B_m \subset U$ and, by regularity, B_n with $x_0 \in B_n$, $\overline{B_n} \subset B_m$; $g_{n,m}$ works.

Step 2. by the imbedding theorem below, $F(x) = (f_1(x), f_2(x), \dots)$ imbeds X in $\mathbb{R}^\omega$ (product topology), which is metrizable (Theorem 2.36). (One may also give a direct metric $D(F(x), F(y)) = \sup|f_n(x) - f_n(y)|/n$.) ∎

Theorem 4.13 (34.2, imbedding theorem)

Let X be a space in which one-point sets are closed and $\{f_\alpha\}_{\alpha \in J}$ continuous functions $X \to \mathbb{R}$ such that for each x_0 and neighbourhood U of x_0 some f_α is positive at x_0 and zero outside U. Then $F(x) = (f_\alpha(x))_{\alpha \in J}$ is an imbedding of X in $\mathbb{R}^J$. If each f_α maps into $[0,1]$, F imbeds X in $[0,1]^J$.

Proof. F is continuous (Theorem 2.31). Injective: for $x \neq y$, choose α with $f_\alpha(x) > 0$ and $f_\alpha = 0$ off $X - \{y\}$ (open since points are closed), so $f_\alpha(y) = 0$. Open onto $Z = F(X)$: let U be open in X and $z_0 = F(x_0) \in F(U)$. Choose β with $f_\beta(x_0) > 0$, $f_\beta = 0$ off U, and let $W = \pi_\beta^{-1}((0,\infty)) \cap Z$, open in Z and containing z_0. If $z = F(x) \in W$, then $f_\beta(x) > 0$, so $x \in U$ and $z \in F(U)$. Thus $F(U)$ is open in Z. ∎

Corollary 4.14 (34.3)

X is completely regular iff it is homeomorphic to a subspace of $[0,1]^J$ for some J.

Proof. $[0,1]^J$ is completely regular (Theorem 4.11), hence so are its subspaces. Conversely, for completely regular X the family of all continuous $f : X \to [0,1]$ satisfies the hypothesis of Theorem 4.13 (separate x_0 from $X - U$). ∎

4.6 The Tietze extension theorem

Motivation. Can a continuous function defined on a closed set be extended to the whole space? In normal spaces it always can; the proof builds the extension as a uniform limit of Urysohn functions.

Theorem 4.15 (35.1, Tietze extension theorem)

Let X be normal and $A \subset X$ closed.
(a) Every continuous $f : A \to [a,b]$ extends to a continuous $g : X \to [a,b]$.
(b) Every continuous $f : A \to \mathbb{R}$ extends to a continuous $g : X \to \mathbb{R}$.

Proof.

Step 1. for continuous $f : A \to [-r, r]$ there is continuous $g : X \to \mathbb{R}$ with $|g| \le r/3$ on X and $|g - f| \le 2r/3$ on A: the sets $B = f^{-1}([-r, -r/3])$, $C = f^{-1}([r/3, r])$ are disjoint and closed in X; Urysohn gives $g : X \to [-r/3, r/3]$ with $g = -r/3$ on B and $r/3$ on C, and one checks the three cases.

Step 2. for $f : A \to [-1, 1]$ apply Step 1 repeatedly: g_1 for f with $r = 1$; g_2 for $f - g_1$ with $r = \frac{2}{3}$; in general g_n with $|g_n| \le \frac{1}{3}(\frac{2}{3})^{n-1}$ and $|f - \sum_{i \le n} g_i| \le (\frac{2}{3})^n$ on A. $g = \sum g_n$ converges uniformly (comparison with a geometric series), so g is continuous (Theorem 2.42), $|g| \le 1$, and $g = f$ on A.

Step 3. for (b), regard f as a map into $(-1, 1) \cong \mathbb{R}$. Extend it by Step 2 to $g : X \to [-1, 1]$. The set $D = g^{-1}(\{-1, 1\})$ is closed and disjoint from A; Urysohn gives $\phi : X \to [0, 1]$ with $\phi = 0$ on D, $\phi = 1$ on A. Then $h = \phi g$ extends f and maps into $(-1, 1)$. ■

Conversely, Tietze's theorem implies the Urysohn lemma (extend the function equal to 0 on A and 1 on B from the closed set $A \cup B$).

4.7 Imbeddings of manifolds

Motivation. Abstract manifolds are defined without reference to any ambient space. Partitions of unity let us embed every compact manifold in some $\mathbb{R}^N$, so abstract manifolds are no more general than submanifolds of Euclidean space.

An **m-manifold** is a Hausdorff space with a countable basis in which each point has a neighbourhood homeomorphic to an open subset of $\mathbb{R}^m$. For $\phi : X \to \mathbb{R}$, the **support** is $\overline{\{\phi \neq 0\}}$. A **partition of unity dominated by** an indexed open covering $\{U_i\}$ is a family of continuous $\phi_i : X \to [0, 1]$ with $\operatorname{supp} \phi_i \subset U_i$ and $\sum \phi_i = 1$.

Theorem 4.16 (36.1, existence of finite partitions of unity)

Every finite open covering $\{U_1, \dots, U_n\}$ of a normal space has a dominated partition of unity.

Proof.

Step 1 (shrinking). there are open V_i covering X with $\overline{V_i} \subset U_i$. Inductively, $A = X - (V_1 \cup \cdots \cup V_{k-1} \cup U_{k+1} \cup \cdots \cup U_n)$ is closed and contained in U_k; choose $V_k \supset A$ with $\overline{V_k} \subset U_k$; the sets $V_1, \dots, V_k, U_{k+1}, \dots, U_n$ still cover.

Step 2. shrink again to W_i with $\overline{W_i} \subset V_i$, and take Urysohn $\psi_i = 1$ on $\overline{W_i}$, 0 off V_i; $\operatorname{supp} \psi_i \subset \overline{V_i} \subset U_i$. $\Psi = \sum \psi_i > 0$ everywhere (the W_i cover), and $\phi_i = \psi_i / \Psi$. ■

Theorem 4.17 (36.2)

Every compact m-manifold imbeds in $\mathbb{R}^N$ for some N.

Proof. Cover X by finitely many open U_i with imbeddings $g_i : U_i \to \mathbb{R}^m$, and let ϕ_i be a dominated partition of unity (X is compact Hausdorff, hence normal). Define $h_i = \phi_i g_i$ on U_i and 0 off $\operatorname{supp} \phi_i$ (continuous by the pasting lemma for open sets), and $F = (\phi_1, \dots, \phi_n, h_1, \dots, h_n) : X \to \mathbb{R}^{n(m+1)}$. F is continuous; injective: if $F(x) = F(y)$, choose i with $\phi_i(x) > 0$; then $\phi_i(y) = \phi_i(x) > 0$, so $x, y \in U_i$, and $\phi_i(x) g_i(x) = \phi_i(y) g_i(y)$ gives $g_i(x) = g_i(y)$, so $x = y$. A continuous injection of a compact space into a Hausdorff space is an imbedding (Theorem 3.20). ■